

6	(a) e.m.f. = total energy available (per unit charge) some (of the available energy) is used/lost/wasted/given out in the internal resistance of the battery (hence p.d. available less than e.m.f.)	B1	B1	[2]
(b)	(i) $V = IR$ $I = 6.9 / 5.0 = 1.4$ (1.38) A	C1	A1	[2]
	(ii) $r = \text{lost volts} / \text{current}$ $= (9 - 6.9) / 1.38 = 1.5(2) \Omega$	C1	A1	[2]
(c)	(i) $P = EI$ (not $P = VI$ if only this line given or 9 V not used in second line) $= 9 \times 1.38 = 12$ (12.4) W	C1	A1	[2]
	(ii) efficiency = output power / total power $= VI / EI = 6.9 / 9$ or $(9.52) / (12.4) = 0.767 / 76.7\%$	C1	A1	[2]

Page 4	Mark Scheme	Syllabus	Paper
	GCE AS/A LEVEL – October/November 2013	9702	23